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LBNet: Linearized Bregman algorithm-based deep unfolding network for hyperspectral image unmixing

Guo-Liang Han, Jie Huang *, Sheng Shu, Wen Yan, Hong-Ji Xie

School of Mathematical Sciences, University of Electronic Science and Technology of China, Chengdu, Sichuan, 611731, PR China

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ABSTRACT

Hyperspectral image unmixing is essential for material identification in remote sensing but faces a fundamental dilemma: model-based methods are interpretable yet sensitive to handcrafted optimization, while deep learning approaches are adaptable but often lack transparency. To bridge this gap, we propose a linearized Bregman algorithm-based deep unfolding network (LBNet) for blind unmixing. LBNet uniquely integrates the interpretability of the linearized Bregman algorithm (LBA) with the adaptability of data-driven learning. Our work makes three key contributions. First, we develop an improved linearized Bregman algorithm for blind unmixing (ILBA-BU), which enables alternating and joint updates of both the endmember and abundance matrices within the LBA framework, extending the traditional LBA from sparse vector recovery to the blind hyperspectral unmixing problem. Second, by unfolding the ILBA-BU into a deep network, we design the LBNet with two interpretable, alternating sub-modules (EndNet and AbuNet). Third, we introduce a novel weighted reconstruction loss that prioritizes crucial spectral bands. Experimental results on both synthetic and real-world datasets demonstrate that LBNet achieves superior endmember estimation and competitive abundance estimation compared with several state-of-the-art methods. The code is available at https://github.com/glianghan/LBNet/tree/main.

1. Introduction

With the continuous development of remote sensing technology, hyperspectral images (HSIs) provide abundant surface information about the earth. These images are widely applied in various fields, including agriculture and water resources management [1], food quality and safety [2], and medical diagnosis [3]. However, their interpretation remains challenging due to environmental complexity. HSIs also support numerous applications such as classification [4], anomaly detection [5], and image fusion [6]. Due to the limited spatial resolution of hyperspectral sensors, a single pixel often contains multiple materials, resulting in mixed pixels. Hyperspectral unmixing (HU) decomposes each pixel in HSIs into constituent materials (endmembers) and their abundances, which represent the fractional contribution of each material to the pixel spectrum [7].

It is well-known that HU is an ill-posed inverse problem centered on recovering endmembers and their abundances from HSIs [8]. Among various HU models, the linear mixing model (LMM) [9] is distinguished by its simplicity, efficiency, and capability to accurately represent the spectral mixing process. LMM assumes that the observed spectrum of a mixed pixel is a linear combination of the endmember spectra weighted by their corresponding abundances. Building on the LMM, researchers propose several effective unmixing algorithms, roughly categorized into geometric, sparse regression, and statistical approaches.

Among geometric methods, vertex component analysis (VCA) [10] is commonly used for endmember extraction. For abundance estimation, fully constrained least squares (FCLS) [11] estimates abundances under the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC). Within sparse regression methods, sparse unmixing via variable splitting and augmented Lagrangian (SUnSAL) [12] leverages the sparsity of abundances to enhance spectral unmixing performance. In statistical methods, nonnegative matrix factorization (NMF) [13] provides an effective framework for HU, as its decomposition is well aligned with the physical characteristics of spectral mixtures and provides good interpretability. These model-based methods offer strong interpretability as they are derived from well-established physical models and mathematical formulations. Nevertheless, current approaches exhibit two critical limitations. First, endmember extraction and abundance estimation are typically performed in sequential stages, which may overlook their inherent interdependence. Second, these methods often require careful parameter tuning [14].

* Corresponding author.
E-mail addresses: glianghan@163.com (G.-L. Han), huangjie_uestc@uestc.edu.cn (J. Huang), shusheng012@126.com (S. Shu), sevenandneves@163.com (W. Yan), hjxie_945@163.com (H.-J. Xie).

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0165-1684/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

In recent years, the rise of deep learning has led to the rapid development of various methods based on deep neural networks in the field of HU, effectively capturing the spectral features of HSIs [15]. Foundational contributions include autoencoder-based approaches [16], with subsequent advancements such as the untied denoising autoencoder with sparsity (uDAS) [17], the hyperspectral unmixing using deep image prior (UnDIP) [18], and the minimum simplex convolutional network (MiSiCNet) [19]. Recent work has also extended deep learning to nonlinear unmixing scenarios, including the deep autoencoder-based augmented network (DAAN) [20] and the extended multilinear mixing model-inspired dual-stream network (EMLM-net) [21]. Although deep learning-based methods have achieved promising performance in HU, their black-box nature often leads to limited interpretability, and their performance may depend on the quality and distribution of the training data.

Deep unfolding methods have recently attracted considerable attention for constructing interpretable neural networks [22,23]. By transforming iterative inference algorithms into networks with a fixed number of layers, these approaches combine the interpretability of model-based optimization with the representation capability of deep learning [24]. These ideas have also been introduced into HU. For example, unfolding networks derived from optimization algorithms such as the iterative shrinkage-thresholding algorithm (ISTA) and the alternating direction method of multipliers (ADMM) have been developed for HU [25,26]. In addition, SNMF-Net unfolds the sparse NMF optimization process into a deep architecture, enabling learnable parameters while preserving the interpretability of the underlying model [27]. Xie and Huang proposed a sparse and total variation (TV) regularization based deep alternating neural network (STV-ANet), which unfolds an alternating optimization algorithm with sparse and TV priors for HU [28]. Zhao et al. proposed an unrolled plug-and-play network (PnP-Net), which unfolds an ADMM-based optimization algorithm and incorporates both internal priors learned from hyperspectral data and external priors from pretrained denoisers [29]. These studies demonstrate that integrating model-based optimization with deep learning provides an effective paradigm for HU. Therefore, the choice of the underlying optimization algorithm plays a key role in determining the properties and performance of the resulting network. In addition to deep unfolding approaches, other approaches have also been explored. For example, sparsity-enhanced convolutional decomposition introduces a 3D convolutional sparse representation to capture spatial–spectral correlations in hyperspectral data, providing an alternative paradigm for blind HU [30].

The linearized Bregman algorithm (LBA) is an iterative optimization method known for its fast convergence and effectiveness, which has been widely applied in various fields such as image deblurring [31], compressed sensing [32], and curvelet reconstruction [33]. These properties suggest that LBA is a promising candidate for HU. Numerous variants of LBA have been developed to enhance its performance, including nonstationary variants that improve computational efficiency while preserving sparsity [34], as well as acceleration techniques such as the Barzilai-Borwein line search and the limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) method [35]. Classical gradient descent methods have also been utilized to enhance LBA's overall efficiency [36]. Despite LBA's success in these applications, to the best of our knowledge, it has not yet been explored for blind HU, which motivates us to investigate its potential for this problem.

In this paper, we propose an improved linearized Bregman algorithm for blind unmixing (ILBA-BU). Specifically, we have extended the classical LBA for abundance estimation. The proposed ILBA-BU is formulated under the ANC and the ASC, and derives update rules for both endmembers and abundances. This formulation enables the joint estimation of endmembers and abundances within the LBA framework. However, ILBA-BU involves multiple parameters requiring careful tuning. To mitigate this complexity, ILBA-BU is unfolded into a deep unfolding network termed LBNet for blind unmixing. In this vein, the LBNet comprises two sub-modules: EndNet and AbuNet, which are designed to update endmembers and abundances, respectively. Through strategically alternating predictions, LBNet effectively leverages the knowledge embedded in the model, leading to more accurate endmember and abundance estimates. In addition, we introduce a weighted reconstruction loss to emphasize informative spectral bands and improve spectral reconstruction accuracy. Notably, LBNet exhibits high interpretability, achieving satisfactory outcomes within just a few dozen iterations while utilizing only a fraction of the pixels. Furthermore, the network parameters are learned in an end-to-end manner, offering significant advantages in both computational efficiency and estimation accuracy.

The main contributions of this paper are as follows:

1) An improved linearized Bregman algorithm for blind HU, termed ILBA-BU, is developed. The proposed algorithm is formulated under the LMM with the ANC and the ASC, enabling the joint estimation of endmembers and abundances. Explicit update rules are derived, providing a theoretical connection between linearized Bregman iteration and blind HU.
2) A deep unfolding network, termed LBNet, is constructed by unfolding the proposed ILBA-BU algorithm into a learnable architecture. LBNet consists of two sub-modules, EndNet and AbuNet, corresponding to endmember and abundance updates, respectively. This design preserves the interpretability of the underlying optimization framework.
3) A weighted reconstruction loss is introduced to emphasize informative spectral bands, which improves spectral reconstruction accuracy and enhances training stability. Specifically, the weighting matrix is derived from the model structure and adaptively adjusts the contribution of different spectral bands during training.

The organization of this article is structured as follows. Section 2 introduces the L_p sparsity-based unmixing model and LBA. In Section 3, we derive the ILBA-BU and the network architecture of LBNet, presenting its learning and implementation details. Section 4 presents experimental results on synthetic and real-world datasets, including ablation studies and comparative evaluations. Finally, Section 5 concludes the article.

2. Related work

2.1. Linear mixture model

The LMM assumes that each pixel spectrum is a linear combination of endmember spectra weighted by their abundances. Using the matrix notation, the LMM can be written as:

$$\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S} + \boldsymbol{E}, \tag{1}$$

where $\boldsymbol{X} \in \mathbb{R}^{L\times N}$ is the observed HSI with L bands and N pixels, $\boldsymbol{A} \in \mathbb{R}^{L\times R}$ is the endmember matrix with each column representing a material spectrum, $\boldsymbol{S} \in \mathbb{R}^{R\times N}$ is the abundance matrix, representing the proportions of materials in each pixel, and $\boldsymbol{E} \in \mathbb{R}^{L\times N}$ denotes noise. To ensure physically meaningful unmixing, two constraints [37] are typically added: the ANC, i.e., $\boldsymbol{S} \geq 0$, and the ASC, i.e., $\mathbf{1}_R^T\boldsymbol{S} = \mathbf{1}_N^T$, where $\mathbf{1}_R \in \mathbb{R}^R$ and $\mathbf{1}_N \in \mathbb{R}^N$ denote column vectors of all ones.

2.2. L_p-NMF for blind unmixing

The sparsity property of the abundance matrix plays a crucial role in enhancing the unmixing accuracy for HU [38–40]. The L_p-NMF model [41] introduces an L_p norm sparsity constraint ($0 < p \leq 1$) on the abundance matrix to improve unmixing performance. The corresponding unmixing model can be formulated as:

$$\begin{aligned} \min_{\boldsymbol{A},\boldsymbol{S}} \ & \mathcal{L}_{\mathrm{NMF}}(\boldsymbol{A},\boldsymbol{S}) = \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_F^2 + \lambda\|\boldsymbol{S}\|_p^p, \\ & s.t.\ \ \boldsymbol{A},\boldsymbol{S} \geq 0, \quad \mathbf{1}_R^T\boldsymbol{S} = \mathbf{1}_N^T, \end{aligned} \tag{2}$$

where $\|\cdot\|_F$ represents the Frobenius norm of a matrix, $\|\boldsymbol{S}\|_p^p = \sum_{r=1}^{R}\sum_{n=1}^{N}|s_{rn}|^p$ promotes the sparsity of $\boldsymbol{S}$, s_{rn} is the (r, n)-th element of $\boldsymbol{S}$, and λ is a nonnegative balancing parameter that controls the trade-off between the data fidelity term and the sparsity regularization. The iterative update rules for the endmember and abundance matrices are given in [27] as follows:

$$\begin{cases} \boldsymbol{A}_{k+1} = \varphi(\boldsymbol{A}_k - t_1(\boldsymbol{A}_k\boldsymbol{S}_k - \boldsymbol{X})\boldsymbol{S}_k^T), \\ \boldsymbol{S}_{k+1} = \zeta(\varphi(\eta_{(\lambda,p)}(\boldsymbol{S}_k - t_2\boldsymbol{A}_{k+1}^T(\boldsymbol{A}_{k+1}\boldsymbol{S}_k - \boldsymbol{X})))), \end{cases} \tag{3}$$

where k denotes the iteration number, t_1 and t_2 represent the step sizes that control the update speeds of $\boldsymbol{A}$ and $\boldsymbol{S}$, respectively. $\varphi(\cdot)$ represents the rectified linear unit (ReLU) activation function [42], which ensures the nonnegativity of $\boldsymbol{A}$ and $\boldsymbol{S}$. $\zeta(\cdot)$ represents the normalization function that guarantees the ASC. The generalized shrinkage thresholding (GST) operator $\eta_{(\lambda,p)}$ is defined as [43]:

$$\eta_{(\lambda,p)}(y) = \begin{cases} \text{sign}(y)(|y| - \lambda p(|y|)^{p-1}) & \text{if } |y| > \Delta_p^{GST}(\lambda), \\ 0 & \text{otherwise}, \end{cases} \tag{4}$$

where $\text{sign}(\cdot)$ denotes the sign function, and $\Delta_p^{\text{GST}}(\lambda)$ is given by

$$\Delta_p^{\text{GST}}(\lambda) = (2\lambda(1-p))^{\frac{1}{2-p}} + \lambda p(2\lambda(1-p))^{\frac{p-1}{2-p}}. \tag{5}$$

2.3. Linearized Bregman algorithm

Recall that LBA is an efficient iterative method commonly employed to solve ill-posed linear inverse problems of the form:

$$\boldsymbol{g} = \boldsymbol{B}\boldsymbol{f} + \boldsymbol{e}, \tag{6}$$

where $\boldsymbol{g} \in \mathbb{R}^N$ is the observed data, $\boldsymbol{B} \in \mathbb{R}^{N\times N}$ is an ill-conditioned matrix, and $\boldsymbol{e} \in \mathbb{R}^N$ represents noise. Such problems frequently arise in various applications like image deblurring, compressed sensing, and signal reconstruction, where the underlying signal $\boldsymbol{f} \in \mathbb{R}^N$ needs to be recovered from noisy or incomplete measurements. LBA aims to recover $\boldsymbol{f}$ by minimizing the following objective function:

$$\mathcal{L}_{\text{LBA}}(\boldsymbol{f}) = \frac{1}{2}\|\boldsymbol{B}\boldsymbol{f} - \boldsymbol{g}\|_{\boldsymbol{D}}^2 + \mu\|\boldsymbol{f}\|_1, \tag{7}$$

where $\boldsymbol{D} = (\boldsymbol{B}\boldsymbol{B}^* + \alpha\boldsymbol{I})^{-1}$ with $\alpha > 0$, and $\mu > 0$ is a regularization parameter controlling the sparsity level. Here, $\boldsymbol{B}^*$ represents the conjugate transpose (or Hermitian transpose) of $\boldsymbol{B}$. $\|\boldsymbol{x}\|_{\boldsymbol{D}}^2 = \langle \boldsymbol{x}, \boldsymbol{D}\boldsymbol{x}\rangle$ holds for any vector $\boldsymbol{x} \in \mathbb{R}^N$ and $\boldsymbol{D} \in \mathbb{R}^{N\times N}$, where $\langle\cdot,\cdot\rangle$ represents the standard inner product. $\|\cdot\|_1$ denotes the L_1 norm of a vector. The iterative update scheme for the modified LBA is given by [34]:

$$\begin{cases} \boldsymbol{z}_{k+1} = \boldsymbol{z}_k + \boldsymbol{B}^*\boldsymbol{D}(\boldsymbol{g} - \boldsymbol{B}\boldsymbol{f}_k), \\ \boldsymbol{f}_{k+1} = S_\mu(\boldsymbol{z}_{k+1}), \end{cases} \tag{8}$$

where $\boldsymbol{z}_k$ denotes the linearized Bregman auxiliary variable, and $S_\mu(\cdot)$ represents the soft-thresholding operator [44], defined as $S_\mu(\boldsymbol{x}) = \text{sign}(\boldsymbol{x})\max(|\boldsymbol{x}| - \mu, 0)$. The goal of the LBA is to compute a sparse solution $\boldsymbol{f}$ that approximates $\boldsymbol{g}$, while mitigating the effects of $\boldsymbol{e}$. By iteratively updating the solution using the Bregman distance, the LBA ensures that the solution remains computationally efficient and sparse during optimization, making it a powerful tool in high-dimensional data analysis and signal processing.

3. Proposed algorithm

In this section, we propose a linearized Bregman algorithm-based deep unfolding framework for blind HU. Unlike prior works that apply linearized Bregman iteration to neural network training or generic inverse problems, we develop a new iterative scheme tailored specifically for HU under the LMM. By unfolding the proposed iterations, we construct a deep network termed LBNet, which preserves the interpretability of the optimization process while introducing learnable parameters to improve estimation performance.

3.1. ILBA-BU

In hyperspectral scenes, each pixel is typically composed of only a few materials, which leads to a sparse structure in the abundance matrix. To exploit this property, we adopt the L_p norm ($0 < p \le 1$) as the sparsity regularization term. Compared with the L_1 norm, the L_p norm promotes stronger sparsity and improves the separability of abundance coefficients. Meanwhile, HU aims to reconstruct the observed spectra as linear combinations of endmembers. In addition, to measure the reconstruction consistency between the observed spectra and the estimated mixtures, we introduce a weighted quadratic fidelity term $\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_{\boldsymbol{P}}^2$, where the weighting matrix is defined as $\boldsymbol{P} = (\boldsymbol{A}\boldsymbol{A}^T + \alpha\boldsymbol{I})^{-1}$ with $\alpha > 0$. This symmetric positive-definite matrix $\boldsymbol{P}$ accounts for spectral correlations induced by the endmember matrix and improves numerical stability when $\boldsymbol{A}\boldsymbol{A}^T$ is ill-conditioned.

Based on the above considerations, we propose a novel blind HU model as follows:

$$\begin{aligned} \min_{\boldsymbol{A},\boldsymbol{S}}\ \mathcal{L}(\boldsymbol{A},\boldsymbol{S}) =& \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_{\boldsymbol{P}}^2 + \lambda\|\boldsymbol{S}\|_p^p, \\ & s.t.\ \boldsymbol{A},\boldsymbol{S} \ge 0, \quad \boldsymbol{1}_R^T\boldsymbol{S} = \boldsymbol{1}_N^T. \end{aligned} \tag{9}$$

where $\|\boldsymbol{S}\|_p^p = \sum_{r=1}^{R}\sum_{n=1}^{N}|s_{rn}|^p$ denotes the L_p norm with $0 < p \le 1$, which promotes sparse abundance representations. The parameter $\lambda > 0$ controls the strength of the sparsity regularization. $\boldsymbol{P} = (\boldsymbol{A}\boldsymbol{A}^T + \alpha\boldsymbol{I})^{-1}$ is the weighting matrix used in the reconstruction fidelity term, where $\alpha > 0$ is a small constant ensuring the positive definiteness of $\boldsymbol{P}$.

To solve the optimization problem in Eq. (9), we adopt the alternating least squares (ALS) method due to its faster convergence rate and ease of implementation [45]. This approach decomposes the optimization into two subproblems, aimed at updating $\boldsymbol{A}$ and $\boldsymbol{S}$, respectively.

- **Update A.** Specifically, at the $k+1$-th iteration, by fixing $\boldsymbol{S}_k$, the optimization problem in Eq. (9) can be reformulated to focus exclusively on optimizing $\boldsymbol{A}$:

$$\min_{\boldsymbol{A}}\ \mathcal{L}(\boldsymbol{A},\boldsymbol{S}_k) = \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_{\boldsymbol{P}}^2, \quad s.t.\ \boldsymbol{A} \ge 0, \tag{10}$$

For this optimization problem, since the objective becomes difficult to differentiate with respect to $\boldsymbol{A}$ after incorporating the weighting matrix $\boldsymbol{P} = (\boldsymbol{A}\boldsymbol{A}^T + \alpha\boldsymbol{I})^{-1}$, we establish a relationship between the weighted Frobenius norm and the Frobenius norm for the matrix approximation $\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k$, where $\boldsymbol{P}$ is a positive-definite matrix.

Recall that $\boldsymbol{P} = (\boldsymbol{A}\boldsymbol{A}^T + \alpha\boldsymbol{I})^{-1}$ is symmetric positive-definite with $\alpha > 0$. Expanding the weighted Frobenius norm yields

$$\begin{aligned} \|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_{\boldsymbol{P}}^2 &= \text{tr}((\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k)^T\boldsymbol{P}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k)) \\ &= \text{tr}((\boldsymbol{P}^{\frac{1}{2}}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k))^T(\boldsymbol{P}^{\frac{1}{2}}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k))) \\ &= \|\boldsymbol{P}^{\frac{1}{2}}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k)\|_F^2. \end{aligned} \tag{11}$$

Applying the compatibility of matrix norms yields

$$\|\boldsymbol{P}^{\frac{1}{2}}(\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k)\|_F \le \|\boldsymbol{P}^{\frac{1}{2}}\|_2\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_F, \tag{12}$$

where $\|\cdot\|_2$ denotes the spectral norm [46]. Let $\lambda_{\min}$ denote the smallest singular value of $\boldsymbol{A}\boldsymbol{A}^T$, then clearly we have $\lambda_{\min} \geqslant 0$. It follows that the largest singular value of $(\boldsymbol{A}\boldsymbol{A}^T + \alpha\boldsymbol{I})^{-1}$ is $\frac{1}{\lambda_{\min}+\alpha}$. Thus

$$\|\boldsymbol{P}^{\frac{1}{2}}\|_2 = \sqrt{\|(\boldsymbol{A}\boldsymbol{A}^T + \alpha\boldsymbol{I})^{-1}\|_2} = \frac{1}{\sqrt{\lambda_{\min} + \alpha}} \leqslant \frac{1}{\sqrt{\alpha}}. \tag{13}$$

Considering Eqs. (11)–(13), we obtain

$$\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_{\boldsymbol{P}} \le \frac{1}{\sqrt{\alpha}}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_F. \tag{14}$$

For computational simplicity, we approximate $\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_{\boldsymbol{P}}$ by $\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_F$ in Eq. (10), leading to the following optimization problem

$$\min_{\boldsymbol{A}}\ \mathcal{L}(\boldsymbol{A},\boldsymbol{S}_k) = \frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}_k\|_F^2, \quad s.t.\ \boldsymbol{A} \ge 0. \tag{15}$$

Solving Eq. (15) via the proximal gradient algorithm [47] yields

$$\boldsymbol{A}_{k+1} = \varphi(\boldsymbol{A}_k - t_1(\boldsymbol{A}_k\boldsymbol{S}_k - \boldsymbol{X})\boldsymbol{S}_k^T), \tag{16}$$

where $t_1 \geq 0$ is the step size, which controls the update speed of $\boldsymbol{A}$.

- **Update S.** With $\boldsymbol{A}_{k+1}$ fixed, the update of $\boldsymbol{S}$ is considered. Let $\boldsymbol{P}_{k+1} = (\boldsymbol{A}_{k+1}\boldsymbol{A}_{k+1}^T + \alpha\boldsymbol{I})^{-1}$. Then the $\boldsymbol{S}$-subproblem of Eq. (9) can be written as:

$$\begin{aligned} \min_{\boldsymbol{S}} \ \mathcal{L}(\boldsymbol{A}_{k+1}, \boldsymbol{S}) = &\frac{1}{2}\|\boldsymbol{X} - \boldsymbol{A}_{k+1}\boldsymbol{S}\|^2_{\boldsymbol{P}_{k+1}} + \lambda\|\boldsymbol{S}\|_p^p, \\ &s.t.\ \boldsymbol{S} \geq 0, \quad \mathbf{1}_R^T\boldsymbol{S} = \mathbf{1}_N^T, \end{aligned} \tag{17}$$

where $\|\boldsymbol{C}\|^2_{\boldsymbol{P}} = \mathrm{tr}(\boldsymbol{C}^T\boldsymbol{P}\boldsymbol{C})$ for any $\boldsymbol{C} \in \mathbb{R}^{L\times N}$ and $\boldsymbol{P} \in \mathbb{R}^{L\times L}$.

This subproblem corresponds to a weighted sparse abundance estimation problem under the LMM. Inspired by the linearized Bregman framework in Eq. (8) for the L_1-norm regularized problem in Eq. (7), we propose the following LBA-based iterative scheme for solving the L_p-norm regularized optimization problem in Eq. (17)

$$\begin{cases} \boldsymbol{Z}_{k+1} = \boldsymbol{Z}_k - t_2\boldsymbol{A}_{k+1}^T\boldsymbol{P}_{k+1}(\boldsymbol{A}_{k+1}\boldsymbol{S}_k - \boldsymbol{X}), \\ \boldsymbol{S}_{k+1} = \zeta\big(\varphi(\eta_{(\lambda,p)}(\boldsymbol{Z}_{k+1}))\big), \end{cases} \tag{18}$$

where $\boldsymbol{Z}_k$ is the auxiliary variable introduced by the linearized Bregman iteration, $t_2 \geq 0$ denotes the step size, $\varphi(\cdot)$ and $\zeta(\cdot)$ correspond to the operators introduced in Eq. (3), $\eta_{(\lambda,p)}(\cdot)$ denotes the GST operator defined in Eq. (4). We note that the proposed updates in Eq. (18) combine the efficiency of the LBA with the enhanced sparsity-promoting capability of the L_p-norm regularization, leading to more discriminative abundance estimates.

Algorithm 1 Pseudocode of ILBA-BU.

Input: $\boldsymbol{X}$.
Initialization: $\boldsymbol{A}_0$, $\boldsymbol{S}_0$, $\boldsymbol{Z}_0$.
Selected parameters: t_1, t_2, λ, p, and α.
for $k = 0, 1, 2, \ldots, 100$
$\quad \boldsymbol{A}_{k+1} = \varphi(\boldsymbol{A}_k - t_1(\boldsymbol{A}_k\boldsymbol{S}_k - \boldsymbol{X})\boldsymbol{S}_k^T)$,
$\quad \boldsymbol{P}_{k+1} = (\boldsymbol{A}_{k+1}\boldsymbol{A}_{k+1}^T + \alpha\boldsymbol{I})^{-1}$,
$\quad \boldsymbol{Z}_{k+1} = \boldsymbol{Z}_k - t_2\boldsymbol{A}_{k+1}^T\boldsymbol{P}_{k+1}(\boldsymbol{A}_{k+1}\boldsymbol{S}_k - \boldsymbol{X})$,
$\quad \boldsymbol{S}_{k+1} = \zeta(\varphi(\eta_{(\lambda,p)}(\boldsymbol{Z}_{k+1})))$.
end for
Output: $\hat{\boldsymbol{A}} = \boldsymbol{A}_{k+1}$, $\hat{\boldsymbol{S}} = \boldsymbol{S}_{k+1}$.

Up until now, we have derived an improved linearized Bregman-based blind unmixing (ILBA-BU) algorithm and established the corresponding iterative update rules for jointly estimating endmembers and abundances. The pseudocode of the proposed ILBA-BU is summarized in Algorithm 1. We note that the algorithm involves several parameters, including t_1, t_2, λ, p, and α, which influence the convergence behavior and estimation performance. Compared with traditional NMF-based unmixing methods (e.g., $L_{1/2}$-NMF and L_p-NMF), the proposed ILBA-BU introduces a weighting matrix $\boldsymbol{P}$ and an auxiliary variable $\boldsymbol{Z}$. These components provide a more flexible optimization structure while preserving the physical interpretability of the LMM, which also facilitates the subsequent deep unfolding design in the next section.

3.2. LBNet

Although the ILBA-BU provides an effective iterative framework for blind HU, it involves multiple parameters that require careful tuning. To address this limitation, we unroll the ILBA-BU and construct a deep unfolding network termed LBNet, which automatically learns the parameters and improves the efficiency and accuracy of HU. LBNet alternately optimizes endmembers and abundances through two sub-modules, **EndNet** and **AbuNet**. As illustrated in Fig. 1, the architecture of LBNet consists of these two key components.

EndNet for endmember estimation. With the computed abundance matrix $\boldsymbol{S}_k$ as input, EndNet iteratively optimizes $\boldsymbol{A}$ through a learning-based approach. In our implementation, the step size parameter t_1 in Eq. (16) is replaced by a learnable network parameter $t_{1,k}$, enabling adaptive control of the optimization step magnitude. This formulation yields the following iterative update rule for endmember refinement:

$$\boldsymbol{A}_{k+1} = \varphi(\boldsymbol{A}_k - t_{1,k}(\boldsymbol{A}_k\boldsymbol{S}_k - \boldsymbol{X})\boldsymbol{S}_k^T). \tag{19}$$

AbuNet for abundance estimation. AbuNet operates with a fixed endmember matrix $\boldsymbol{A}_{k+1}$ and iteratively optimizes $\boldsymbol{S}$. We initialize $\boldsymbol{Z}_0 = \boldsymbol{S}_0$ and replace α in $\boldsymbol{P}_{k+1}$ with a network parameter α_k, obtaining $\boldsymbol{P}_{k+1} = (\boldsymbol{A}_{k+1}\boldsymbol{A}_{k+1}^T + \alpha_k\boldsymbol{I})^{-1}$. The optimization objective, as shown in Eq. (17), quantifies both the reconstruction error of the HSI data and enforces sparsity constraints on $\boldsymbol{S}$. The parameters t_2, λ, and p in Eq. (18) are implemented as learnable network parameters $t_{2,k}$, λ_k, and p_k. Recall that in Algorithm 1, the term $\boldsymbol{A}_{k+1}^T\boldsymbol{P}_{k+1}$ serves as a regularized projection from a spectral space to an endmember space. However, directly parameterizing this operator would tightly couple the network parameters with the current estimate of $\boldsymbol{A}_{k+1}$, reducing the flexibility of the learned representation. To enable adaptive representation learning while preserving the algorithmic structure, we introduce a learnable matrix $\boldsymbol{W}_k \in \mathbb{R}^{R\times L}$ and adopt the structured operator $\boldsymbol{W}_k(\boldsymbol{W}_k^T\boldsymbol{W}_k + \alpha_k\boldsymbol{I})^{-1}$ to approximate the projection term $\boldsymbol{A}_{k+1}^T\boldsymbol{P}_{k+1}$. When $\boldsymbol{W}_k$ approaches $\boldsymbol{A}_{k+1}^T$, the resulting operator recovers the original projection behavior. The regularization term $\alpha_k\boldsymbol{I}$ ensures numerical stability during matrix inversion. Thus, the abundance updates follow:

$$\begin{cases} \boldsymbol{Z}_{k+1} = \boldsymbol{Z}_k - t_{2,k}\boldsymbol{W}_k(\boldsymbol{W}_k^T\boldsymbol{W}_k + \alpha_k\boldsymbol{I})^{-1}(\boldsymbol{A}_{k+1}\boldsymbol{S}_k - \boldsymbol{X}), \\ \boldsymbol{S}_{k+1} = \zeta(\varphi(\eta_{(\lambda_k,p_k)}(\boldsymbol{Z}_{k+1}))), \end{cases} \tag{20}$$

where $\boldsymbol{W}_k \in \mathbb{R}^{R\times L}$ denotes the parameter matrix and $t_{2,k}$ represents the step size.

Loss function. In conventional deep HU methods, the mean squared error (MSE) loss $\|\boldsymbol{X} - \boldsymbol{A}\boldsymbol{S}\|_F^2$ is commonly adopted to measure the spectral reconstruction error. However, this formulation treats all spectral components equally and ignores the spectral correlations induced by the endmember matrix. To better reflect the reconstruction structure in the proposed model, we adopt a weighted quadratic loss derived from the linearized Bregman formulation. Specifically, the reconstruction error is measured by the weighted norm $\|\boldsymbol{X} - \boldsymbol{A}_k\boldsymbol{S}_k\|^2_{\boldsymbol{P}_k}$, where $\boldsymbol{P}_k = (\boldsymbol{A}_k\boldsymbol{A}_k^T + \bar{\alpha}\boldsymbol{I})^{-1}$ with $\bar{\alpha} > 0$. This weighting matrix captures the spectral correlations introduced by the endmember matrix and improves numerical stability when $\boldsymbol{A}_k\boldsymbol{A}_k^T$ is ill-conditioned.

Based on this formulation, the training loss of LBNet is defined as:

$$\mathcal{L}_\theta = \frac{1}{KN}\sum_{k=1}^{K}\|\boldsymbol{X} - \boldsymbol{A}_k\boldsymbol{S}_k\|^2_{(\boldsymbol{A}_k\boldsymbol{A}_k^T + \bar{\alpha}\boldsymbol{I})^{-1}}, \tag{21}$$

where $\theta = \{t_{1,k}, t_{2,k}, \boldsymbol{W}_k, \alpha_k, \lambda_k, p_k\}$ denotes the learnable parameters and K is the number of network layers. The scalar $\bar{\alpha} > 0$ is introduced to ensure the positive definiteness of the weighting matrix and to stabilize gradient propagation during training.

To facilitate implementation and reproducibility, the overall workflow of the proposed LBNet is summarized below. Specifically, the network alternately updates endmembers and abundances through the two sub-modules **EndNet** and **AbuNet** across K unfolding layers. In each layer, the learnable parameters are optimized via backpropagation based on the weighted reconstruction loss in Eq. (21). This training strategy enables the network to adaptively learn the parameters while preserving the algorithmic structure derived from ILBA-BU.

4. Experiments

In this section, we compare LBNet with seven state-of-the-art blind unmixing methods: uDAS [17], UnDIP [18], MiSiCNet [19], CyCU-Net [48], ADMMNet [26], UST-Net [49], and SNMF-Net [27]. Among these

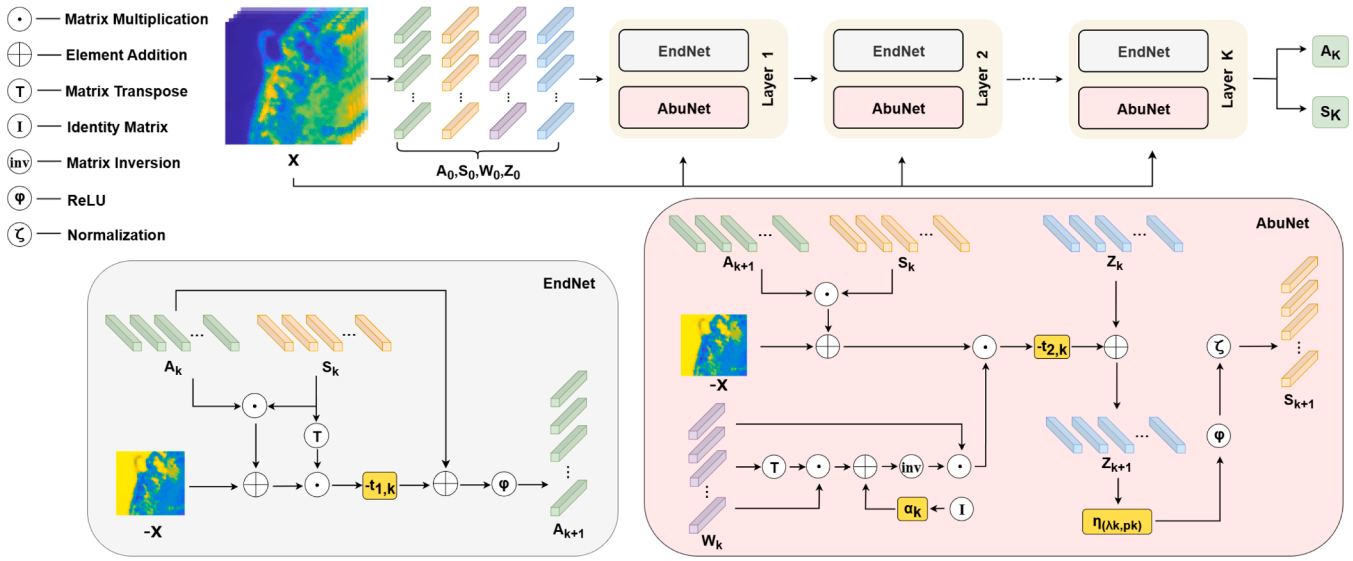


Fig. 1. The framework of the proposed LBNet consists of two principal sub-modules, EndNet and AbuNet, which are responsible for updating $\boldsymbol{A}$ and $\boldsymbol{S}$, respectively. The significance of the symbols has been delineated in the accompanying figure. All parameters highlighted in yellow are learnable and are optimized during the training process.

methods, uDAS, UnDIP, MiSiCNet, CyCU-Net, and UST-Net are learning-based methods, while ADMMNet and SNMF-Net are unfolding-based methods similar to LBNet.

All experiments are conducted on a workstation equipped with an Intel(R) Core(TM) i9-9900K CPU and an NVIDIA GeForce RTX 4070 GPU. To ensure a fair comparison, all algorithms are implemented using the optimal parameter settings as reported in the corresponding literature, and using identical initial endmembers and consistent initialization strategies whenever applicable. Additionally, each algorithm is tested ten times, and the average results are reported to enhance the reliability of the findings. Both SNMF-Net and LBNet are trained using identical training samples for a fair comparison. For LBNet implementation, $\boldsymbol{A}_0$ and $\boldsymbol{S}_0$ are initialized using VCA and FCLS, respectively. $\boldsymbol{W}_0$ is obtained through generalized mutual coherence (GMC) minimization [27]. Initialized $t_{1,0}$ and $t_{2,0}$ are set as $1/(\|\boldsymbol{S}_0\boldsymbol{S}_0^T\|_2^2+\epsilon)$ and $1/(\|\boldsymbol{A}_0^T\boldsymbol{A}_0\|_2^2+\epsilon)$ with $\epsilon=0.1$, respectively. λ_0, p_0, α_0, $\overline{\alpha}$, and K are set to $0.01\times t_{2,0}$, 0.5, 1, 0.01, and 24, respectively. The maximum number of iterations is set to 100.

We employ the spectral angle distances (SAD) and the root mean squared error (RMSE) as performance evaluation metrics. Generally, lower values of SAD and RMSE indicate better unmixing performance. Let $\boldsymbol{A}$ and $\hat{\boldsymbol{A}}$ represent the reference and estimated endmember matrices, respectively. Let $\boldsymbol{S}$ and $\hat{\boldsymbol{S}}$ denote the reference and estimated abundance maps, respectively. The mathematical formulas for the SAD and the RMSE of the r-th materials are as follows:

$$\mathrm{SAD}_r = \arccos\frac{\boldsymbol{a}_r^T\hat{\boldsymbol{a}}_r}{\|\boldsymbol{a}_r\|_2\|\hat{\boldsymbol{a}}_r\|_2}, \quad \mathrm{RMSE}_r = \sqrt{\frac{1}{N}\sum_{n=1}^{N}(\hat{s}_{rn}-s_{rn})^2}, \quad (22)$$

where $\boldsymbol{a}_r$ denotes the r-th column of $\boldsymbol{A}$, for $r=1,\cdots,R$.

4.1. Simulated experiments

We conduct experiments on the synthetic dataset with a resolution of 64×64 pixels. Six spectral signatures are selected from the U.S. Geological Survey (USGS) spectral library[1], including Carnallite, Ammonio-jarosite, Almandine, Brucite, Axinite, and Chlonte. These signatures contain 224 spectral bands with wavelengths ranging from 0.38 to 2.5 μm,

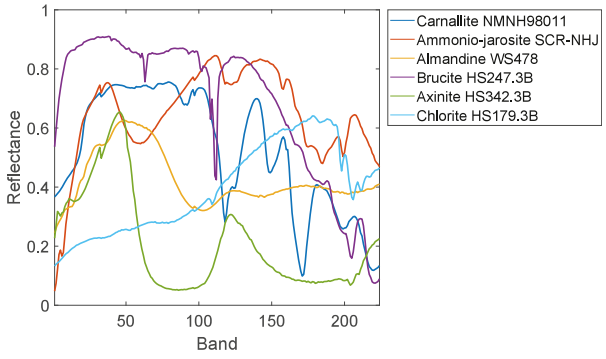


Fig. 2. Spectral signatures of six endmembers used in the synthetic dataset.

and their spectral profiles are shown in Fig. 2. The corresponding abundance matrices are generated following the procedure in [50]. Specifically, each pixel is first constructed as a mixture of two endmembers rather than a single pure signature. Subsequently, spatial Gaussian filtering is applied to introduce additional mixing among neighboring pixels. The observed data are generated using the LMM and then contaminated with white i.i.d. Gaussian noise with signal-to-noise ratio (SNR) of 15, 20, and 25 dB, respectively. For training, 1000 pixels are randomly selected.

Table 1 reports the SAD, RMSE, their mean values, and running time for different methods under SNR = 15, 20, and 25 dB. At SNR = 15 and 20 dB, LBNet achieves the lowest mean SAD and mean RMSE among all compared methods, and attains the best (or second-best) scores for most endmembers. At SNR = 25 dB, SNMF-Net achieves the best mean SAD and mean RMSE, while LBNet ranks second in terms of both metrics with competitive performance.

Figs. 3 and 4 provide qualitative comparisons at SNR = 20 dB. LBNet produces abundance maps with fewer noise artifacts and clearer spatial structures. The estimated endmember spectra are also closer to the references compared with most competing methods. Among the unfolding-based baselines, ADMMNet and SNMF-Net exhibit larger discrepancies for some endmembers, while LBNet maintains more consistent reconstruction quality.

Fig. 5 depicts the convergence trajectory of the proposed LBNet. We see from Fig. 5 that the SAD values decrease steadily as the iterations

[1] Available online: http://speclab.cr.usgs.gov/spectral.lib06.

Table 1
Quantitative performance comparison on the synthetic dataset with SAD, RMSE, their mean values, and runtime (seconds) by different unmixing methods. Best and second-best results are displayed in bold and italics, respectively.

SNR	Criteria		uDAS	UnDIP	MiSiCNet	CyCU-Net	ADMMNet	UST-Net	SNMF-Net	LBNet
15dB	SAD	End #1	**0.0273**	0.0500	0.1210	0.2284	0.0818	0.1787	0.0528	*0.0299*
		End #2	**0.0229**	0.0726	0.0857	0.1047	0.0739	0.1813	*0.0354*	**0.0229**
		End #3	0.0589	0.0679	0.1216	0.2760	0.1216	0.3122	**0.0550**	*0.0586*
		End #4	**0.0149**	0.0723	0.0919	0.0763	0.0238	0.0267	0.0422	*0.0184*
		End #5	0.0926	0.1649	0.4410	0.4436	0.1101	**0.0435**	0.1567	*0.0512*
		End #6	0.0695	0.1223	0.2670	0.2726	*0.0549*	0.2638	0.0682	**0.0327**
		Mean	*0.0477*	0.0917	0.1880	0.2336	0.0777	0.1677	0.0684	**0.0356**
	RMSE	End #1	*0.0597*	0.0688	0.1175	0.3012	0.0801	0.1234	0.0687	**0.0555**
		End #2	**0.0521**	0.0851	0.1368	0.2761	0.0566	0.1340	0.0594	*0.0533*
		End #3	0.0888	0.1291	0.1647	0.3112	0.1054	0.1679	*0.0838*	**0.0792**
		End #4	**0.0436**	0.0661	0.1556	0.2350	0.0620	0.1739	0.0616	*0.0477*
		End #5	0.0601	0.0749	0.1734	0.4441	*0.0442*	0.0534	0.0515	**0.0430**
		End #6	0.0629	0.0872	0.1522	0.4611	*0.0502*	0.1274	0.0559	**0.0432**
		Mean	*0.0612*	0.0852	0.1500	0.3381	0.0664	0.1300	0.0635	**0.0537**
	Runtime		**2.52**	208.39	34.76	*12.49*	31.23	81.59	31.45	23.57
SNR	Criteria		uDAS	UnDIP	MiSiCNet	CyCU-Net	ADMMNet	UST-Net	SNMF-Net	LBNet
20dB	SAD	End #1	*0.0156*	0.0374	0.1210	0.1687	0.0335	0.1337	0.0251	**0.0132**
		End #2	**0.0104**	0.0630	0.0861	0.1343	0.0517	0.0795	0.0184	*0.0126*
		End #3	0.0342	0.0807	0.1220	0.2781	0.0650	0.2566	**0.0165**	*0.0274*
		End #4	*0.0081*	0.0449	0.0917	0.0696	0.0171	0.0171	0.0165	**0.0046**
		End #5	*0.0934*	0.1437	0.4418	0.4473	0.0974	0.0646	0.1099	**0.0554**
		End #6	0.0527	0.1091	0.2670	0.2585	0.0609	0.0788	*0.0321*	**0.0318**
		Mean	*0.0357*	0.0798	0.1883	0.2261	0.0543	0.1051	0.0364	**0.0242**
	RMSE	End #1	*0.0349*	0.0645	0.1185	0.3049	0.0460	0.1132	0.0394	**0.0319**
		End #2	0.0355	0.0678	0.1375	0.2634	0.0428	0.0641	*0.0354*	**0.0342**
		End #3	0.0593	0.0721	0.1623	0.3079	0.0722	0.1454	**0.0471**	*0.0527*
		End #4	*0.0299*	0.0478	0.1544	0.2356	0.0392	0.1550	0.0329	**0.0289**
		End #5	0.0558	0.0597	0.1736	0.4405	*0.0316*	0.0463	**0.0300**	0.0371
		End #6	0.0436	0.0603	0.1513	0.4277	0.0389	0.0684	*0.0347*	**0.0316**
		Mean	0.0432	0.0620	0.1496	0.3300	0.0451	0.0987	*0.0366*	**0.0361**
	Runtime		**2.85**	203.65	34.23	*12.65*	35.24	84.68	33.02	23.90
SNR	Criteria		uDAS	UnDIP	MiSiCNet	CyCU-Net	ADMMNet	UST-Net	SNMF-Net	LBNet
25dB	SAD	End #1	0.0307	0.0569	0.1210	0.1348	0.0687	0.1367	**0.0141**	*0.0237*
		End #2	0.0169	0.0605	0.0862	0.0545	0.0147	0.0500	*0.0124*	**0.0114**
		End #3	*0.0158*	0.0762	0.1221	0.2088	0.0834	0.2827	0.0160	**0.0147**
		End #4	0.0104	0.0701	0.0920	0.0895	*0.0065*	0.0262	0.0117	**0.0047**
		End #5	0.1173	0.1691	0.4419	0.4387	*0.0660*	0.0699	**0.0386**	0.0759
		End #6	0.0658	0.1229	0.2676	0.2488	*0.0387*	0.0626	**0.0099**	0.0459
		Mean	0.0428	0.0926	0.1885	0.1959	0.0463	0.1047	**0.0171**	*0.0294*
	RMSE	End #1	0.0306	0.0595	0.1179	0.3178	0.0584	0.1147	**0.0226**	*0.0247*
		End #2	0.0295	0.0821	0.1363	0.2797	**0.0180**	0.0685	*0.0224*	0.0258
		End #3	0.0531	0.0885	0.1619	0.3040	0.0526	0.1700	**0.0277**	*0.0445*
		End #4	0.0334	0.0611	0.1540	0.2344	0.0590	0.1746	**0.0231**	*0.0238*
		End #5	0.0423	0.0631	0.1731	0.4344	*0.0169*	0.0645	**0.0162**	0.0263
		End #6	0.0454	0.0716	0.1514	0.4226	*0.0243*	0.0864	**0.0189**	0.0283
		Mean	0.0391	0.0710	0.1491	0.3322	0.0382	0.1131	**0.0218**	*0.0289*
	Runtime		**2.81**	211.37	34.67	*12.45*	34.56	86.36	31.87	23.94

proceed and gradually stabilize after approximately 80 iterations. This behavior indicates that the proposed network can achieve stable convergence under different noise levels, demonstrating the effectiveness of the proposed unfolding design.

4.2. Real-data experiments

To validate the effectiveness of LBNet, experiments were conducted on four widely used hyperspectral datasets: the Samson [51], Apex [52], Orchard [53], and Cuprite datasets [54]. We note that for the Samson and Apex datasets, reference endmembers and abundances from [55,56] are used for evaluation. For the Orchard dataset, the provided reference endmembers and abundances are used for evaluation [57]. For the Cuprite dataset, only reference endmembers are available, and therefore the SAD is used as the evaluation metric [48].

1) **Samson**: The dataset contains 95×95 pixels with 156 bands covering wavelengths from 401 to 889 nm. The scene consists of three main materials: soil, tree, and water. For model training, 1000 pixels are randomly selected from the dataset.
2) **Apex**: The dataset is a cropped image containing 111×122 pixels with 285 bands spanning wavelengths from 413 to 2420 nm. Four reference endmembers are manually selected from the HSI: road, tree, roof, and water. For model training, 1000 pixels are randomly selected from the dataset.
3) **Orchard**: The dataset contains 20×20 pixels with 216 spectral bands and three endmembers: tree, weed, and soil. One invalid band is removed before evaluation. For model training, 400 pixels are used due to the limited number of available samples.
4) **Cuprite**: The dataset corresponds to the well-known AVIRIS Cuprite scene with a size of 250×190 pixels and 188 spectral bands after removing bands affected by water vapor and low SNR. The scene

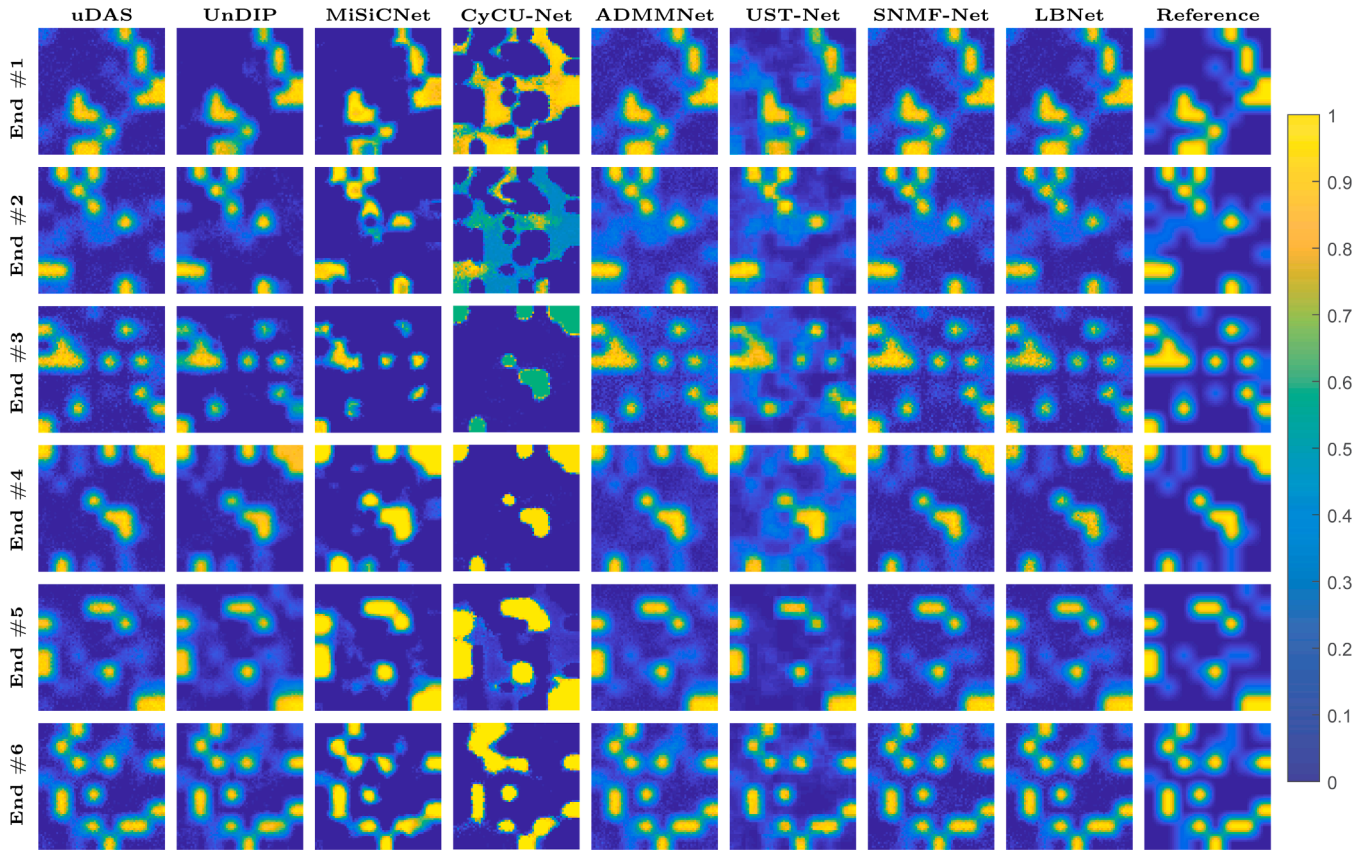


Fig. 3. Comparison of reference and estimated abundance maps by different unmixing algorithms for the synthetic dataset with SNR = 20 dB.

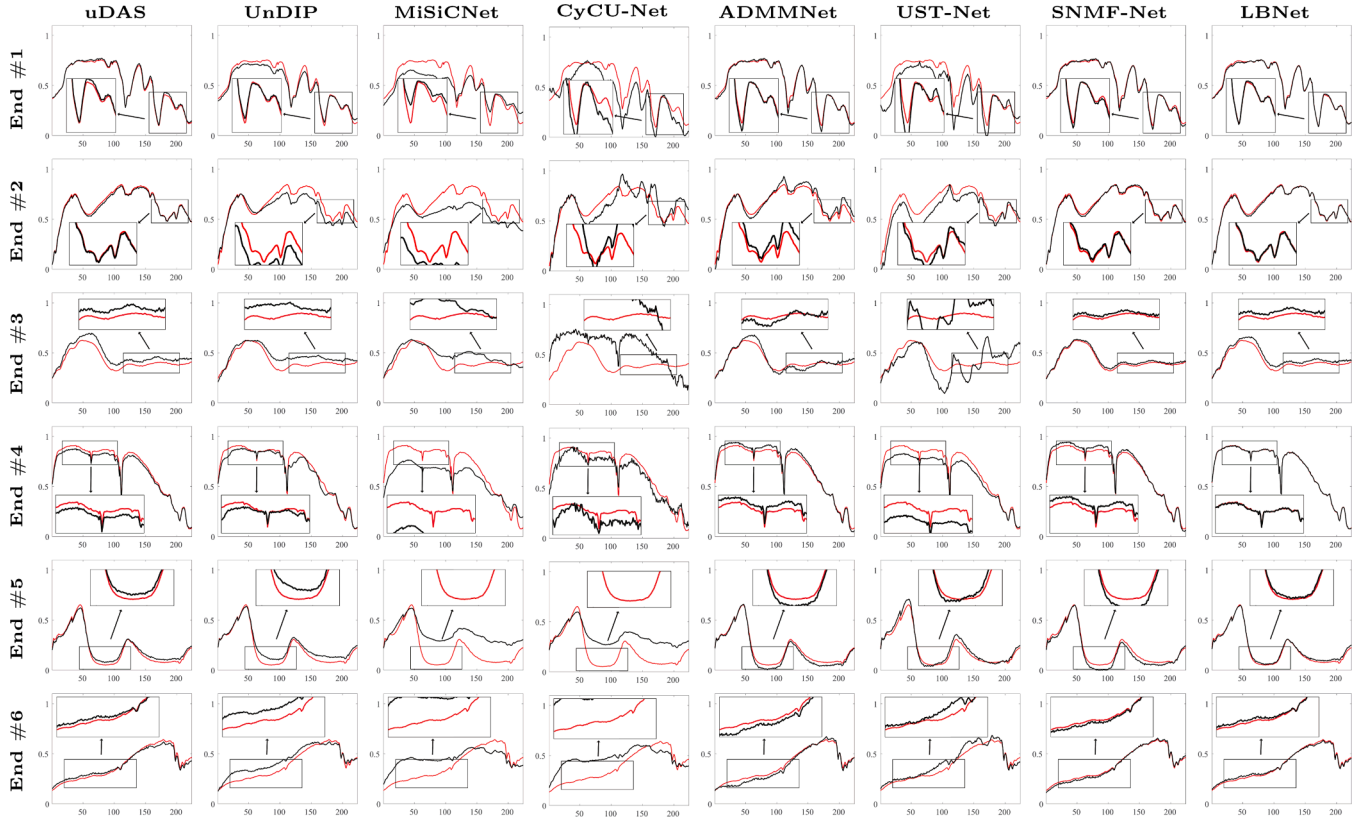


Fig. 4. Comparison of reference and estimated endmembers by different unmixing algorithms for the synthetic dataset with SNR = 20 dB. (*Red: reference endmembers. Black: estimated endmembers.*).

Table 2
SAD and RMSE values of different unmixing algorithms on the Samson and Apex datasets. Best and second-best results are displayed in bold and italics, respectively.

			uDAS	UnDIP	MiSiCNet	CyCU-Net	ADMMNet	UST-Net	SNMF-Net	LBNet
Samson	SAD	Soil	0.0096	0.0127	**0.0076**	0.0655	0.6085	0.1415	0.0128	*0.0084*
		Tree	0.0108	**0.0082**	0.0265	0.0154	0.0116	*0.0087*	0.0106	0.0089
		Water	0.1825	0.1307	0.3135	0.1644	0.1827	0.1853	*0.0994*	**0.0454**
		Mean	0.0676	0.0505	0.1159	0.0818	0.2676	0.1118	*0.0409*	**0.0209**
	RMSE	Soil	0.0472	*0.0347*	0.0414	0.3831	0.2432	0.1316	0.0360	**0.0337**
		Tree	0.0276	**0.0201**	0.0584	0.4709	0.2342	0.1329	*0.0226*	0.0284
		Water	0.0997	*0.0200*	0.0378	0.4723	0.0236	0.0859	0.0202	**0.0125**
		Mean	0.0582	*0.0249*	0.0458	0.4421	0.1670	0.1168	0.0263	**0.0248**
Apex	SAD	Road	1.2910	**0.0908**	*0.2606*	0.4666	0.6639	0.1250	0.7789	0.7204
		Tree	*0.1142*	0.1339	0.1376	0.2484	0.1751	0.1335	0.2027	**0.0903**
		Roof	0.1295	*0.0689*	0.1762	0.0861	0.1047	0.1268	0.2074	**0.0572**
		Water	0.3499	0.5083	0.7670	0.3275	*0.1912*	**0.0508**	0.3644	0.2952
		Mean	0.4711	*0.2005*	0.3354	*0.2821*	0.2837	**0.1090**	0.3883	0.2908
	RMSE	Road	0.1873	0.1742	**0.1366**	0.2328	0.1828	*0.1727*	0.2387	0.2089
		Tree	**0.0816**	0.2154	0.1492	0.4851	0.2469	*0.1311*	0.1325	0.1322
		Roof	0.1295	0.2678	**0.1138**	0.3923	0.2342	0.1282	0.1198	*0.1186*
		Water	*0.1272*	0.4264	0.1984	0.4934	0.4186	0.1517	0.1832	**0.1189**
		Mean	**0.1314**	0.2709	0.1495	0.4009	0.2706	0.1459	0.1685	*0.1447*

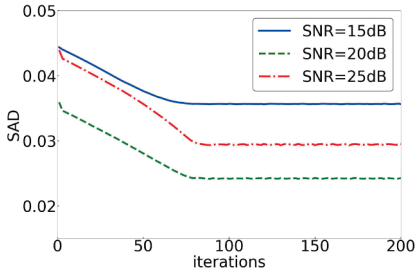


Fig. 5. Plot of SAD by LBNet against iterations for the synthetic dataset with different SNR levels.

contains multiple exposed minerals. Following [48], we have selected 12 representative minerals to evaluate the unmixing performance, including Muscovite, Alunite, Kaolinite KGa-2 (pxl), Kaolinite KGa-1 (wxl), Kaolin-Smectite KLF506, Nontronite, Buddingtonite, Dumortierite, Kaolin-Smectite H89-FR-5, Montmorillonite, Chalcedony, and Desert Varnish. These minerals are denoted as End #1–End #12. For model training, 5500 pixels are randomly selected from the dataset.

Table 2 reports the quantitative comparison on the Samson and Apex datasets. On the Samson dataset, LBNet achieves the lowest mean SAD and RMSE among all evaluated methods, demonstrating superior unmixing accuracy. In particular, LBNet significantly reduces the SAD for the water endmember compared with the second-best method, while maintaining competitive performance for the soil and tree endmembers. On the Apex dataset, UST-Net obtains the lowest mean SAD, whereas LBNet achieves the best SAD for the tree and roof endmembers. Moreover, the RMSE results of LBNet remain competitive across all materials. Overall, LBNet achieves the best or second-best results for several endmembers and evaluation metrics, indicating robust performance across different real hyperspectral scenes.

To further illustrate the qualitative performance, Figs. 6 and 7 present the abundance maps and endmember spectra obtained on the Samson dataset. The results show that LBNet produces accurate abundance maps and endmember spectra, with clear spatial structures and good agreement with the reference signatures. Similarly, on the Apex dataset, LBNet achieves competitive performance. Although UST-Net obtains the best SAD results, the abundance maps produced by LBNet still exhibit clear spatial structures and reasonable agreement with the reference distributions. Due to space limitations, the corresponding visualization results for the Apex dataset are omitted here.

Table 3 presents the quantitative comparison on the Orchard and Cuprite datasets. On the Orchard dataset, UST-Net achieves the lowest mean SAD, followed by ADMMNet, while LBNet obtains competitive spectral estimation accuracy. For abundance estimation, ADMMNet achieves the lowest mean RMSE, followed by UST-Net, while LBNet provides comparable performance to several other methods. On the Cuprite dataset, LBNet achieves the lowest mean SAD among all compared methods, while most other methods produce larger errors for several endmembers. These results demonstrate that LBNet provides accurate endmember estimation on real hyperspectral scenes.

Table 4 compares the runtime and memory consumption on the four real datasets. We see from Table 4 that LBNet achieves competitive computational efficiency across all datasets. On the Samson and Apex datasets, LBNet is substantially faster than UnDIP and ADMMNet, while CyCU-Net remains the fastest due to its lightweight architecture. For the larger Cuprite dataset, LBNet requires only 43.54 s, significantly outperforming most baseline methods. In terms of memory consumption, LBNet requires more GPU memory than some baseline methods, which is expected due to its multi-stage unfolding architecture. Nevertheless, the memory requirement remains practical for modern GPU hardware.

4.3. Hyperparameter analysis

To determine appropriate hyperparameter settings for LBNet, we now analyze the influence of the number of network layers, the number of training samples, and the initialization parameters on the synthetic dataset with SNR = 25 dB.

Influence of the number of network layers. Figs. 8 (a) and (b) illustrate the effect of the number of network layers on the unmixing performance. The SAD decreases as the number of network layers increases, while RMSE stabilizes beyond 24 layers with only minor variations. When the network depth becomes excessively large, RMSE slightly increases, which may be attributed to potential overfitting caused by the overly complex model. Considering both accuracy and computational efficiency, the number of network layers K is set to 24.

Influence of the number of training samples. Fig 8 (c) and (d) show the performance versus the number of training samples. LBNet achieves near-optimal SAD and RMSE using 1000 training pixels (24.41% of the dataset), indicating that the network can be effectively trained using only a subset of the available samples.

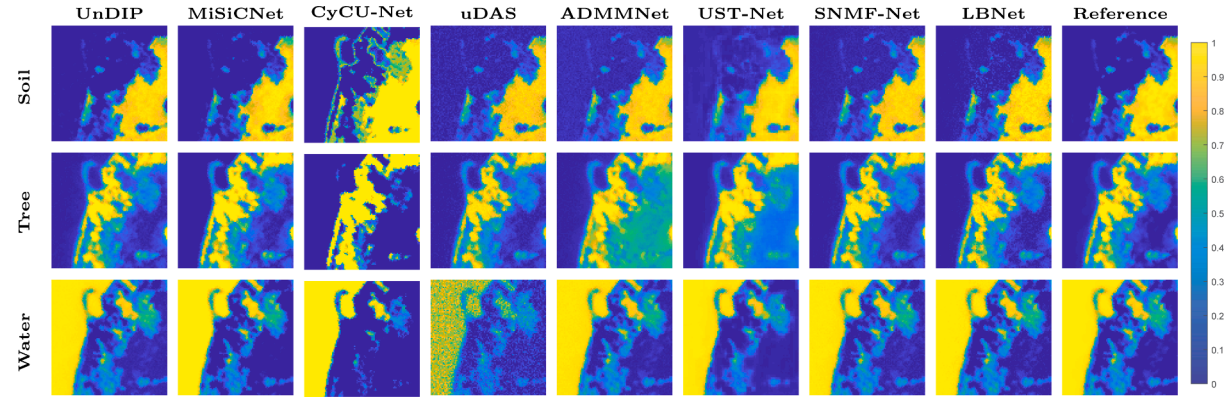


Fig. 6. Comparison of reference and estimated abundance maps by different unmixing algorithms for the Samson dataset.

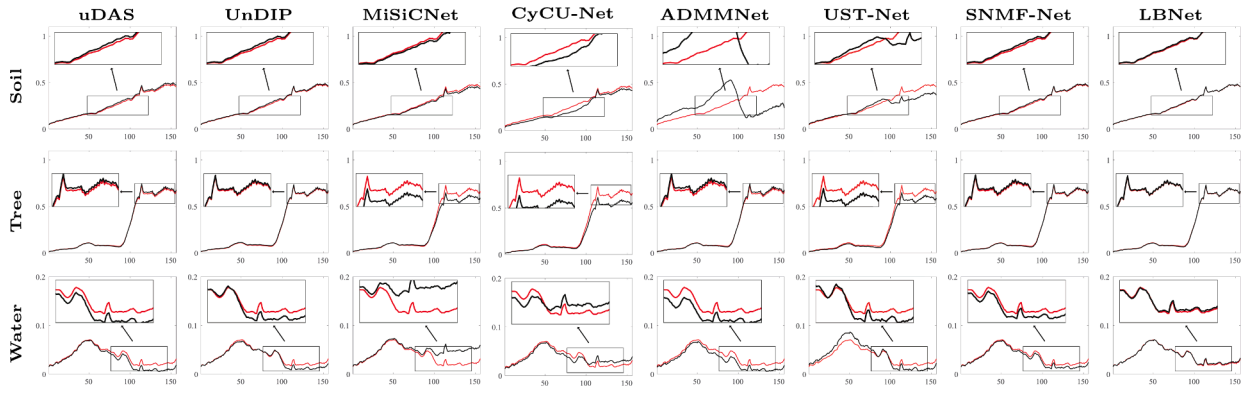


Fig. 7. Comparison of reference and estimated endmembers by different unmixing algorithms for the Samson dataset. (*Red: reference endmembers. Black: estimated endmembers*).

Table 3
Quantitative comparison on the Orchard and Cuprite datasets. For the Orchard dataset, SAD and RMSE are computed w.r.t. the provided reference endmembers and abundance maps. For the Cuprite dataset, only SAD is reported. Best and second-best results are displayed in bold and italics, respectively.

			uDAS	UnDIP	MiSiCNet	CyCU-Net	ADMMNet	UST-Net	SNMF-Net	LBNet
Orchard	SAD	Tree	0.2439	**0.2312**	0.3613	0.2757	0.2365	0.2444	0.2363	*0.2358*
		Weed	0.1762	**0.1213**	*0.1538*	0.2859	0.1650	0.1853	0.1765	0.1539
		Soil	0.1306	0.1563	0.3943	0.2635	*0.1004*	**0.0636**	0.1343	0.1317
		Mean	0.1836	0.1696	0.3031	0.2750	*0.1673*	**0.1644**	0.1824	0.1738
	RMSE	Tree	0.2515	0.3956	0.2643	0.6144	**0.1080**	*0.1301*	0.2762	0.2728
		Weed	0.4742	0.5015	**0.2786**	0.3571	0.2854	*0.2828*	0.5239	0.4723
		Soil	0.2466	**0.1259**	0.2462	0.5424	0.1997	*0.1948*	0.2736	0.2294
		Mean	0.3241	0.3410	0.2630	0.5047	**0.1977**	*0.2026*	0.3579	0.3249
Cuprite	SAD	End #1	0.1500	**0.0844**	0.1153	0.1232	0.4140	0.2396	0.0955	*0.0900*
		End #2	*0.1142*	0.2111	0.1441	0.2308	0.3339	**0.1046**	0.5375	0.1236
		End #3	0.1635	0.1629	0.1438	0.1769	0.4314	0.4083	*0.1339*	**0.1043**
		End #4	0.1152	**0.1051**	0.2072	0.2907	1.0038	0.3234	1.2850	*0.1089*
		End #5	0.1780	0.1651	*0.0886*	0.1893	0.3168	0.3827	1.5402	**0.0704**
		End #6	0.1452	*0.1248*	**0.1172**	0.2566	0.2614	0.2466	1.5585	0.1507
		End #7	*0.0716*	**0.0701**	0.1189	0.1586	0.5440	0.2010	0.0835	0.0821
		End #8	0.1366	0.1683	**0.1168**	0.1479	0.4038	0.2690	0.6916	*0.1179*
		End #9	*0.0710*	0.1412	0.1465	0.1155	0.9755	0.3230	1.5708	**0.0667**
		End #10	0.1996	0.1979	0.1632	**0.1149**	0.2142	0.2265	1.0030	*0.1378*
		End #11	0.1274	*0.1188*	0.1651	0.1771	0.3157	0.3696	0.2151	**0.1052**
		End #12	**0.0922**	0.1419	*0.1122*	0.2551	0.4008	0.2245	1.5708	0.1429
		Mean	*0.1304*	0.1410	0.1366	0.1864	0.4679	0.2766	0.8571	**0.1084**

Influence of initialization parameters p_0 and α_0. Fig. 9 illustrates the sensitivity of LBNet to the initialization parameters. The SAD varies within 0.028–0.038 for $\alpha_0 \in [0.6, 1.5]$ and $p_0 \in [0.1, 1.0]$, while RMSE remains within 0.029–0.033. These results indicate that LBNet is relatively robust to the initialization settings. The parameters ($p_0 = 0.5, \alpha_0 = 1$) lie within the stable region and are therefore adopted in the subsequent experiments.

Overall, these experiments demonstrate that LBNet is relatively robust to the choice of hyperparameters, and stable performance can be achieved with moderate network depth, a limited number of training

Table 4
Runtime (seconds) and peak memory consumption (MB) of different unmixing algorithms on four real datasets. All methods are evaluated using their original implementations. For CPU-based methods, peak memory refers to the incremental resident set size (RSS). For GPU-based methods, peak memory corresponds to the maximum allocated GPU memory.

Dataset	Metric	uDAS	UnDIP	MiSiCNet	CyCU-Net	ADMMNet	UST-Net	SNMF-Net	LBNet
Samson	Runtime (s)	2.81	211.37	33.28	20.85	73.27	118.08	31.87	22.56
	Memory (MB)	44.58	155.77	190.75	159.72	89.28	276.94	80.49	382.33
Apex	Runtime (s)	12.8	1111.91	63.42	9.90	110.72	248.54	32.37	27.08
	Memory (MB)	120.23	248.6	326.83	244.1	138.57	415.61	110.07	603.51
Orchard	Runtime (s)	2.07	26.90	15.69	6.07	4.99	25.98	20.50	25.22
	Memory (MB)	192.30	42.18	43.89	36.20	57.46	203.78	78.95	64.13
Cuprite	Runtime (s)	568.39	1837.49	204.11	159.03	589.21	634.36	150.05	43.54
	Memory (MB)	464.98	718.62	1142	725.06	259.36	628.26	459.09	606.03
Device		CPU	GPU	GPU	CPU	CPU	CPU	CPU	GPU

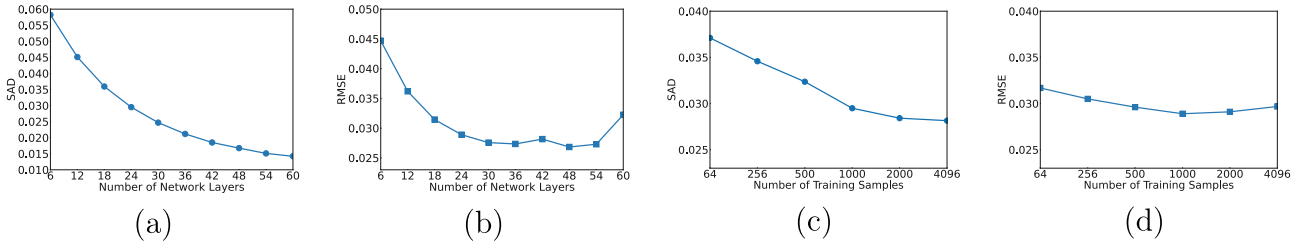


Fig. 8. Influence of the number of network layers and training samples for the synthetic dataset with SNR = 25 dB. (a) SAD and (b) RMSE versus the number of network layers. (c) SAD and (d) RMSE versus the number of training samples.

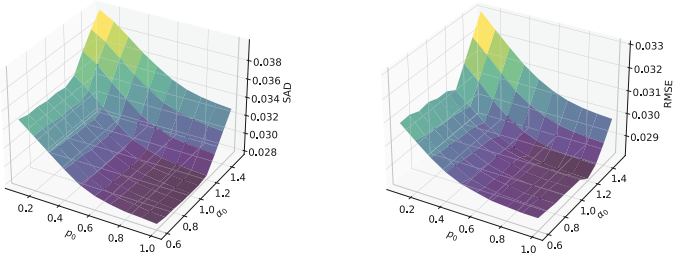


Fig. 9. SAD and RMSE versus the initialization parameters p_0 and α_0 for the synthetic dataset with SNR = 25 dB.

Table 5
Ablation study of LBNet on the synthetic dataset with SNR = 20 dB.

Configuration	Mean SAD	Mean RMSE
Full LBNet	**0.0242**	0.0361
A. Effect of weight matrix $\boldsymbol{P}$		
Weight matrix $\boldsymbol{P}$ ($\boldsymbol{P} = \boldsymbol{I}$)	0.0311	0.0363
Fixed $\boldsymbol{P}$ (Non-learnable)	0.0585	0.0731
B. Effect of parameter learning strategy		
Fixed parameters $t_{1,k}$, $t_{2,k}$, α_k, λ_k, p_k	0.0599	0.0537
Fixed α (Non-learnable)	*0.0251*	**0.0347**
Shared parameters across layers	0.0286	*0.0355*
C. Effect of network architecture & update scheme		
EndNet only (fixed $\boldsymbol{S}$)	0.0580	–
AbuNet only (fixed $\boldsymbol{A}$)	–	0.0439
Sequential update (non-alternating)	0.0707	0.0542

samples, and reasonable initialization settings.

4.4. Ablation study

To evaluate the contribution of the main components in LBNet, we conduct ablation experiments on the synthetic dataset. Unless otherwise specified, the experiments are performed with SNR = 20 dB. The full LBNet baseline corresponds to the complete unfolding network derived from ILBA-BU, where all update modules, auxiliary variables, and learnable parameters are enabled and jointly optimized.

Effect of the weight matrix $\boldsymbol{P}$. The weighting matrix $\boldsymbol{P}$ is introduced to adaptively adjust the contribution of different spectral components in the reconstruction term. To evaluate its impact, we compare the full LBNet with two variants: (1) replacing $\boldsymbol{P}$ with the identity matrix $\boldsymbol{I}$, and (2) using a fixed non-learnable $\boldsymbol{P}$. As reported in Table 5, both variants lead to noticeable performance degradation. In particular, replacing $\boldsymbol{P}$ with $\boldsymbol{I}$ increases the mean SAD from 0.0242 to 0.0311, while fixing $\boldsymbol{P}$ further deteriorates the results. These observations demonstrate that the adaptive weighting mechanism plays an important role in improving spectral reconstruction accuracy.

Effect of parameter learning strategy. LBNet adopts a layer-wise parameter learning strategy that allows each unfolding stage to adaptively adjust its update parameters. To evaluate its effectiveness, we consider several variants, including fixing all parameters, fixing only α, and sharing parameters across layers. As shown in Table 5, fixing all parameters significantly degrades performance, indicating the importance of parameter learning. Using shared parameters across layers also leads to noticeable degradation, demonstrating the advantage of layer-wise parameter adaptation. Fixing only α yields comparable performance to the full LBNet, achieving the best RMSE while producing slightly higher SAD. This observation suggests that although α can remain fixed without severe performance loss, jointly learning all parameters still provides more balanced spectral reconstruction accuracy.

Effect of the $\boldsymbol{Z}$-update scheme. The auxiliary variable $\boldsymbol{Z}$ is introduced in LBNet to mimic the linearized Bregman iteration, which enables a stable abundance update through an inverse-scale-space mechanism. LBNet adopts an ILBA-style update scheme derived from ILBA-BU. To evaluate its effectiveness, we compare the proposed ILBA-style update with two alternative schemes: an ISTA-style update that directly updates $\boldsymbol{S}$ without introducing an auxiliary variable [25], and an ADMM-style update based on a different variable splitting strategy [26]. All other components remain identical to the full LBNet baseline. The quantitative comparison is reported in Table 6. The proposed ILBA-style update achieves the best mean SAD, indicating superior spectral

Table 6
Comparison of different abundance update schemes on the synthetic dataset with SNR = 20 dB. Best and second-best results are displayed in bold and italics, respectively.

Update scheme in LBNet	Mean SAD	Mean RMSE	Runtime (s)
ILBA-style	**0.0242**	*0.0361*	23.90
ISTA-style	0.0319	0.0892	**14.96**
ADMM-style	*0.0307*	**0.0349**	*18.63*

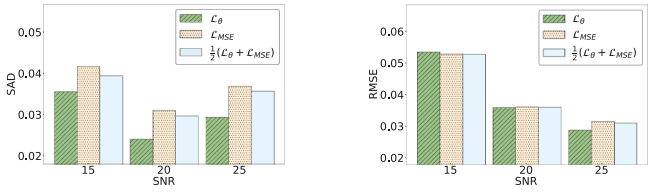


Fig. 10. SAD and RMSE by LBNet for the synthetic dataset with SNR = 15, 20, and 25 dB across different loss functions.

reconstruction accuracy. Although the ADMM-style update attains slightly lower RMSE, its spectral angle accuracy is inferior. In contrast, the ISTA-style update significantly degrades performance, suggesting that directly updating abundances without the Bregman auxiliary variable is less effective for blind hyperspectral unmixing.

Effect of the weighted reconstruction loss. LBNet employs a weighted reconstruction loss to emphasize informative spectral bands during network training. To evaluate its effectiveness, we compare the conventional MSE loss with the proposed weighted reconstruction loss under different noise levels. The results are illustrated in Fig. 10. As shown in the figure, the proposed loss consistently achieves lower SAD while maintaining comparable RMSE across different SNR levels. This indicates that the weighting mechanism improves spectral estimation accuracy by highlighting informative spectral components, thereby enhancing the overall training stability of LBNet.

Overall, the ablation results demonstrate that the main components of LBNet, including the adaptive weighting matrix, the layer-wise parameter learning strategy, the ILBA-style $\boldsymbol{Z}$-update mechanism, and the weighted reconstruction loss, all contribute to the improved unmixing performance.

5. Conclusion

In this article, we have first proposed a novel algorithm ILBA-BU, which extends the LBA model to the problem of blind unmixing, and then developed a deep unfolding network termed LBNet. Particularly, LBNet exhibits strong interpretability, as each network module corresponds to a specific step of the proposed optimization algorithm. All parameters in the network are explainable and support end-to-end training. In addition, a weighted reconstruction loss is introduced to emphasize informative spectral bands and improve training stability. Evaluated on both synthetic and real-world hyperspectral datasets, LBNet demonstrates superior performance compared to state-of-the-art methods. It achieves higher accuracy in estimating endmembers and abundances, while requiring fewer training samples and leveraging domain knowledge more effectively. In our future research, we plan to investigate ways of incorporating spatial information into the LBNet design to further improve HU performance.

CRediT authorship contribution statement

Guo-Liang Han: Writing – original draft, Software, Methodology; **Jie Huang:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition; **Sheng Shu:** Writing – review & editing, Methodology; **Wen Yan:** Software, Methodology; **Hong-Ji Xie:** Software, Methodology.

Data availability

All datasets are publicly available.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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